DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**CHEST TUBE INSERTION**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-PICU-010 (v3)**		
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1.0 PURPOSE

1.1. To formulate standardized criteria for nursing procedure in inserting chest drainage to pediatric patients.

1.2. To re-establish negative intra pleural pressure necessary to lung re-expansion when pneumothorax has developed.

2.0 DEFINITION/ABBREVIATION

2.1. **Chest drainage** is the insertion of tube into the pleural space to evacuate air or fluid, and/or help regain negative pressure.

2.2. **Pneumothorax** is an accumulation of air in the pleural space. Classified as:

2.2.1. **Spontaneous** is a sudden onset of air in the pleural space with deflation of the affected lung in the absence of trauma.

2.2.2. **Open** is an opening in the chest wall large enough to allow air to pass freely in and out of thoracic cavity with each attempted respiration.

2.2.3. **Tension** is a build of air under pressure in the pleural space resulting in interference with filling of both the heart and lungs.

2.3. **Pleural Effusion** is an accumulation of fluid.

2.4. **Hemothorax** is an accumulation of blood.

2.5. **Empyema** is an accumulation of pus.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible in assessing patient's condition before and after chest drainage insertion and writing an order for any procedure and medication needed.

3.2 **Head Nurse** is responsible to make sure that the unit staffs are well knowledge in assisting insertion of chest drainage and all needed equipment are available in the unit for emergency cases.

3.3 **Charge Nurses** are responsible to delegate the procedure to a competent nurse before the procedure and to double check physicians' orders if carried out and help the staff nurse throughout the procedure including double checking medications.

3.4 **Staff nurses** are responsible in carrying out physician's orders, preparing needed equipment before the procedure and assisting physician. It is their responsibility as well to frequently reassess patient condition.

4.0 POLICY

4.1. Insertion of chest drainage shall be performed by a physician with assistance of nurse.

4.2. Aseptic technique shall be observed throughout the procedure.

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4.3. Chest tube shall be monitor for air leak every hour and pleurovac chamber will be monitored for fluctuation every hour.

4.4. Rubber tipped clamps should be ready at bedside.

4.5. Chest tube set shall be readily available for any emergency cases that need urgent intervention.

5.0 PROCEDURE (S)

5.1. Preparatory Phase

5.1.1. Assess patient for pneumothorax, hemothorax, presence of respiratory distress.

5.1.2. Obtain a chest x-ray. Other means of localization of pleural fluid include ultrasound or fluoroscopic localization as order by physician.

5.1.3. Obtain informed consent.

5.1.4. Verify right patient and right location / procedure.

5.1.5. Give Analgesia as ordered.

5.1.6. Assemble drainage system.

5.1.7. Reassure the patient or parents and explain briefly the steps of the procedure.

5.1.8. Position the patient as for an intercostals nerve block or according to physician preference.

5.2. Performance Phase

5.2.1. Needle or intracath technique:

5.2.1.1. Using standard precaution, the skin is prepared, anesthetized and draped, using local anesthetic with a short 25-gauge needle and using aseptic technique. A larger needle is used to infiltrate the subcutaneous tissue, intercostal muscles and parietal pleura.

5.2.1.2. The intracath catheter is inserted through the needle into the pleural space. The needle is removed and the catheter is pushed several centimeters into the pleural space.

5.2.1.3. The catheter is taped to the skin; maybe sutured to the chest wall and covered with a dressing.

5.2.1.4. The catheter is attached to a connector / tubing and attached to a drainage system (underwaterseal or commercial system) and all connections taped.

5.2.1.5. ***Trocar technique***: Using a standard precaution and aseptic technique, a trocar catheter is used for the insertion of a large-bore tube for removal if a moderate to large amount of air leak or for the evacuation of serous effusion.

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5.2.1.6. A small incision is made over the prepared, anesthetized site. Blunt dissection (with a hemostat) through the muscle planes in the interspaces to the parietal pleura is performed.

5.2.1.7. The trocar is directed into the pleural space, the cannula is removed, and a chest tube is inserted into the pleural space and connected to a drainage system. There is a trocar catheter available equipped with an indwelling pointed rod for ease of insertion.

5.2.1.8. The tube is sutured in place and covered with sterile dressing Prevents dislodgment.

5.2.1.9. Catheter is attached to a connector/tube and to the system. All connections are taped. Clamps are removed from the chest tube once connected to the drainage system. Chest tubes open to air at the time of insertion will result in pneumothorax.

5.2.1.10. Take vital signs. Document time of chest tube insertion, name of physician performing the procedure, time of chest x-ray done, initial amount of drainage and patient tolerance to procedure.

5.3. **Follow-up phase**

5.3.1. Observe the drainage system for blood and air. Observe for fluctuation in the tube on respiration.

5.3.2. Secure a follow-up chest x-ray as ordered by physician.

5.3.3. Monitor vital signs every hour or as per physician's order, observing for signs of dyspnea, restlessness, irritability and fever.

5.3.4. Mark the drainage from the collection chamber with tape on the outside of the drainage unit.

5.3.5. Do not elevate drainage set above level of chest. Turn child only from back to side in which tube is placed.

5.3.6. Assess for bleeding, infection, leakage of air and fluid around the tube.

5.3.7. Maintain integrity of the chest drainage system.

6.0 EQUIPMENTS

6.1. Chest Tube set

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

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8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0011(V2) – Chest Tube Insertion – PMSH, 15/02/1439

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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